Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	*Autonomous Production Systems: Utilizing metaverse strategies to increase efficiency and reduce errors in manufacturing*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		*Knowledge Services*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	*Press Machines* *Cutting/processing machinery*	*Human resources capacity development service*	*Human-system interaction technologies*
Concept of Proposed Technology	*Enhance production engineering through AI and digital twins by integrating artificial intelligence and digital twins of production systems and processes to address current challenges in the field. The objective is to optimize the production of complex workpieces for process chains in forming and milling, enabling precise, stage-by-stage control and improved decision-making. This approach tackles issues such as inconsistent quality, unpredictable downtime, and complex process coordination, while reducing downtime and increasing machine utilization, including during unmanned shifts. Autonomous monitoring and predictive maintenance enable significant gains in availability without compromising safety or quality. The project aims to minimize operational complexity and optimize machines through intelligent, autonomous production systems that maintain high batch*			

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	consistency. Real-time visibility into production throughput through digital twins allows rapid adjustments to process parameters and clear visibility of bottlenecks. Interdependencies will be modeled to optimize sequencing, tooling paths, and resource allocation, improving throughput management and reducing constraints. The systems will continuously learn from operation data to progressively enhance performance and reliability over time. Clear diagnostics, faster issue resolution, and reduced cognitive load for operators will be delivered. Ultimately, AI and digital twins will serve as core enablers of efficiency, resilience, and competitiveness in modern manufacturing environments by directly addressing prevailing production engineering challenges.
Rationale for Joint R&D Project	*The machining and the metal forming sector, as part of the wider manufacturing industry, share overarching challenges such as an increasing pressure to operate according to sustainable principles or the technological complexity associated with a higher level of implementation of a digitalized, smart manufacturing environment. (1) Since the introduction of "Industry 4.0" by the German government in 2011 (2), the machining and the metal forming sector have started working on the integration of smart manufacturing methods in their respective processes. As implied by the concept of the fourth industrial revolution, the ultimate goal of this ongoing industrial transformation is the creation of a workerless, fully automated, and digitalized production plant with the capability of intelligently mass-manufacturing customized goods (3, 4). Furthermore, the necessity of this project stems from the growing complexity of modern manufacturing systems, which are increasingly interconnected and data-rich. Without a structured initiative, managing this complexity becomes impractical, as manual approaches cannot keep pace with the dynamic interdependencies across processes, tools, and resources. As manufacturing processes evolve, there is a rising need for advanced analytics, real-time visibility, and predictive decision-making to ensure consistent quality and reliable throughput. In addition, the industry faces a persistent skills gap, with shortages of qualified specialists to monitor, diagnose, and optimize highly automated systems. This talent scarcity amplifies the risk of inefficiencies, longer ramp-up times, and slower response to faults or market changes. Coupled with intensifying competitive pressure, manufacturers must leverage AI, digital twins, and intelligent automation to maintain edge, reduce downtime, and optimize the use of capital equipment.*
Key Words	1. Autonomous Manufacturing 2. Optimal production 3. Digital Twins and Data-driven optimization for process chains

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)
R&D Objectives	ㅇOptimized production of complex workpieces for process chains in the field of forming technology and milling through the use of AI and digital twins of production systems and processes ㅇReduction of downtimes and increase in machine utilization time for unmanned shifts ㅇReducing operational complexity and optimizing machines through the use of autonomous, intelligent production systems			
R&D Target Specifications	ㅇDevelopment of a technology and approach to reduce scrap and production errors (>50% reduction) using data-driven AI-models and industrial metaverse approaches ㅇSystem for training lightweight (and real-time capable) models and digital twins using synthetic data from complex physical models (70% reduction in computing time) ㅇDevelopment of a model- and platform-based technology for the realization and integration of autonomous, smart production technology as an original equipment and retrofit solution ㅇDevelopment of autonomous production systems for monitoring and optimizing (using process control) ongoing production processes without human intervention ㅇDevelopment of technology demonstrators with SMEs as pilot project(s) to evaluate, optimize and introduce the developed technologies			
R&D Plan	Phase 1 – Month 1 to 6 "conceptualization" • Definition of the underlying production technology (machines, processes, etc.) • Analysis of the level of digitization achieved • Screening of challenges in the underlying production scenario • Derivation of useful metaverse components • Methodology for obtaining synthetic data for the production process • Concept for model- and platform-based technology for autonomous production • Concept of the technology demonstrator Phase 2 – Month 7 to 24 „Realization" • Development of models in suitable test environments • Testing of models in the laboratory (at Fraunhofer IWU or at a partner's premises) • Creation of suitable digital twins (using appropriate standardization routines and methods) • Optimization of models and procedures • Development of process control routines Phase 3 – Month 25 to 30 „Demonstration" • Construction of a demonstrator at a Korean SME • Integration of routines, software, and models • Testing and verification of functionality • Finalization of the demonstration scenario and spillover			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	ㅇDevelopment (and further development) and integration of sensor and monitoring technologies as well as control architectures for optimal process control at hardware level in the production of complex workpieces ㅇ Further development of an edge cloud continuum to implement data-driven models and the industrial metaverse in the necessary form for optimizing production and Integration in the pilot scenario(s) ㅇ Creation of precise physical models in the area of process chains for machining and forming technology for the generation of synthetic data for the efficient training of digital twins and lightweight models ㅇ Integration of available, innovative strategies for optimizing production at process chain level and for system and process commissioning ㅇDevelopment and mapping of new, digital and flexible business models for the resulting technical solutions
Expected R&R of Korean SME Partner	ㅇDevelopment of the necessary digital twins using artificial intelligence and platform-based methods for data analysis and interpretation ㅇIntegrating implicit knowledge into autonomous production systems for the optimal and efficient autonomous interpretation of events and making decisions to regulate production ㅇDevelopment of procedures and technologies for the rapid training of lightweight AI models and extended digital twins for autonomous production systems using synthetic data and industrial metaverse approaches ㅇDevelopment and construction of a human-machine interaction interface for easy system optimization and operation of complex production systems without detailed expert knowledge ㅇDevelopment of pilot applications to optimize and evaluate the resulting autonomous and smart production systems ㅇIntegration of the models and software into the pilot applications, optimization and continuation ㅇDevelopment of IP (together with Fraunhofer) and realization in products for industrial application ㅇRealization of the new, digital business models and strategies for the solutions developed within the project

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Technical: ○ Optimized production (higher OEE) ○ Less rejects in the production of complex components ○ Optimized quality with increased productivity ○ Technical solutions for integrating autonomous production systems in the greenfield and brownfield Economical: ○ Cost reduction in the manufacture of complex components ○ Reduction of scrap and machine costs ○ Increased market share in the area of production through increased competitiveness through the use of cutting edge technology and increased export capabilities Social: ○ Reduction of emissions, CO2 and waste ○ Improving working conditions for plant operators and production workers

6. References

(1) R. Kurth et al. Neue Wege der digitalen Verwertung in der Produktionstechnik, Fraunhofer Verlag, 2026, DOI 10.24406/publica-4345

(2) K. Zhou, T. Liu, and L. Zhou, "Industry 4.0: Towards Future Industrial Opportunities and Challenges," in *2015 12th International Conference on Fuzzy Systems and Knowledge Discovery*, Zhangjiajie (China), 2015, pp. 2147–2152. Accessed: Sep. 4 2025. [Online]. Available: https://ieeexplore.ieee.org/document/7382284

(3) W. Xiang, K. Yu, F. Han, Le Fang, D. He, and Q.-L. Han, "Advanced Manufacturing in Industry 5.0: A Survey of Key Enabling Technologies and Future Trends," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 2, pp. 1055–1068, 2024, doi: 10.1109/TII.2023.3274224.

(4) J. Leng *et al.,* "Industry 5.0: Prospect and retrospect," *Journal of Manufacturing Systems*, vol. 65, pp. 279–295, 2022, doi: 10.1016/j.jmsy.2022.09.017.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization